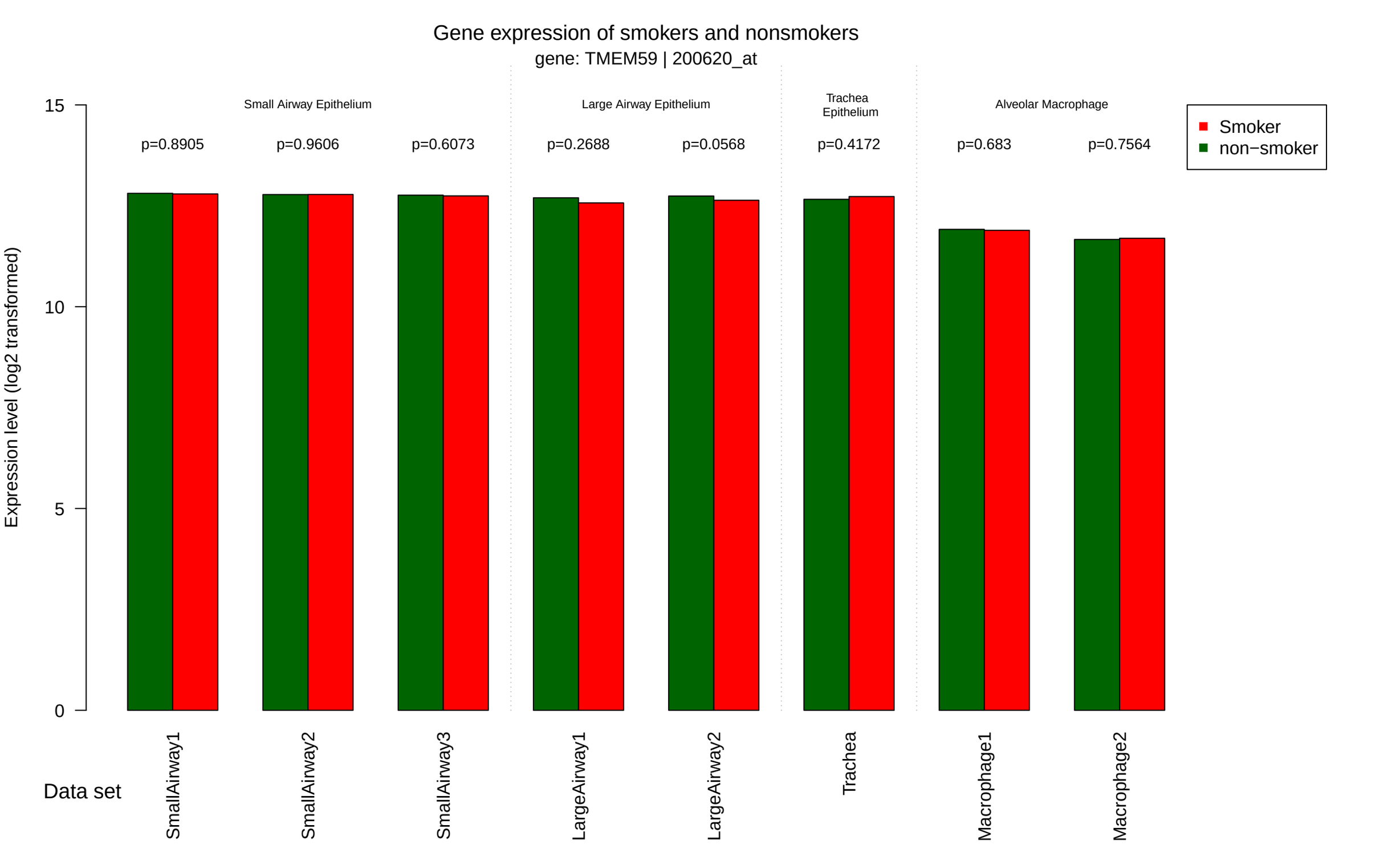
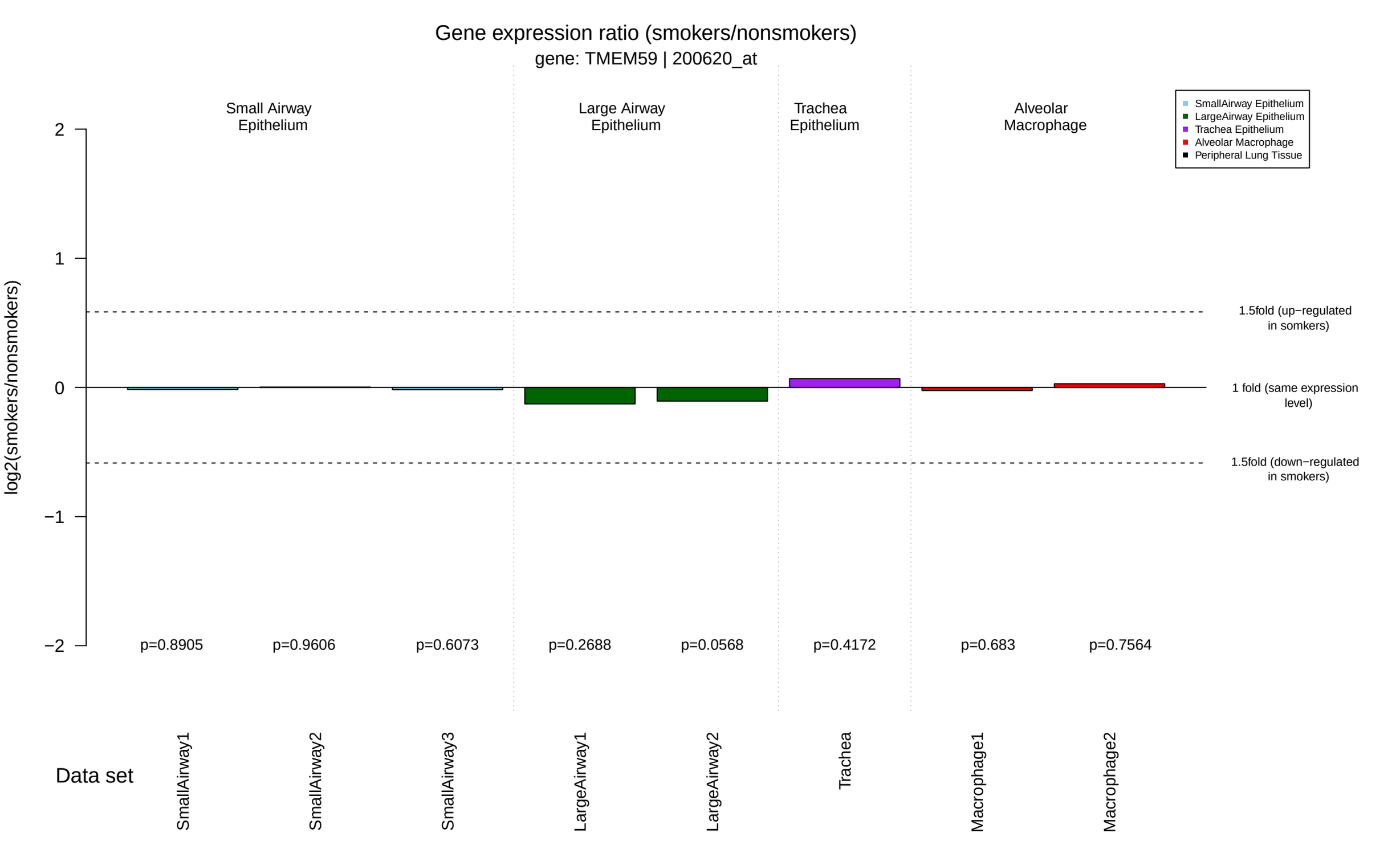
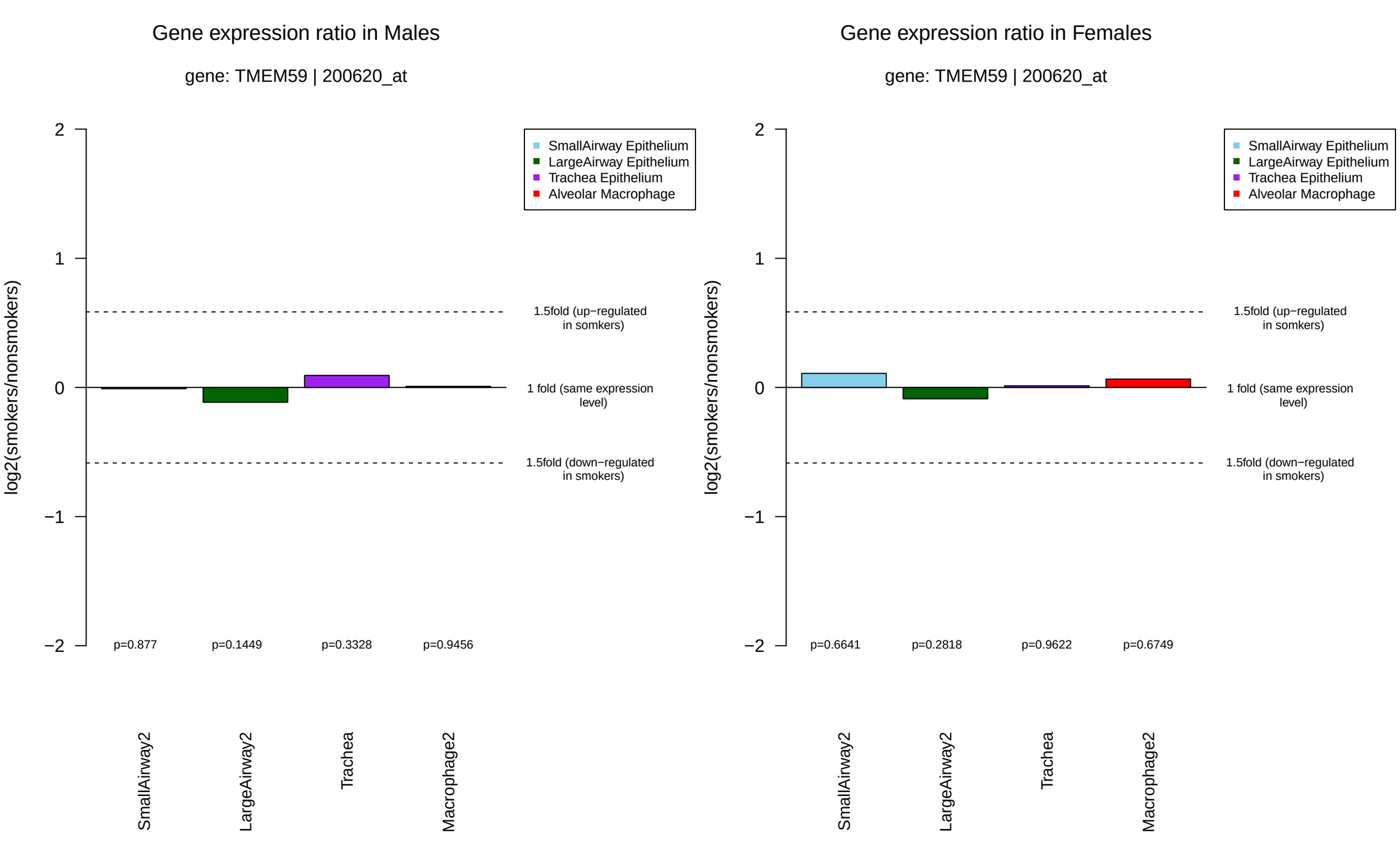
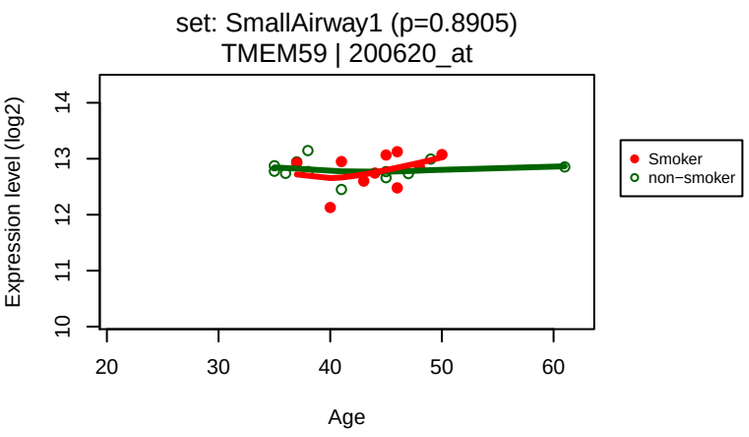
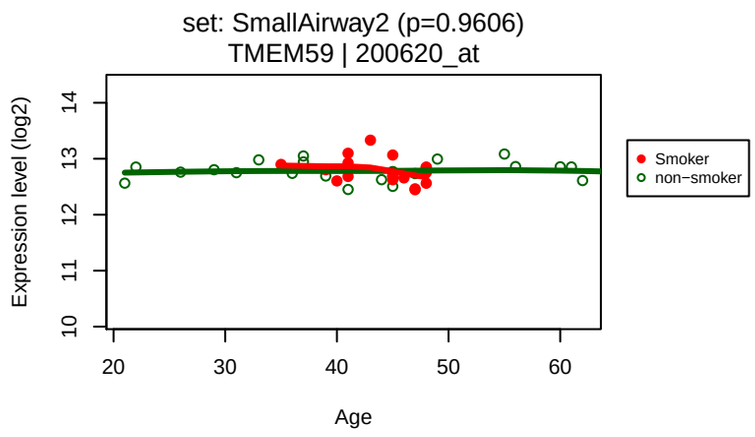
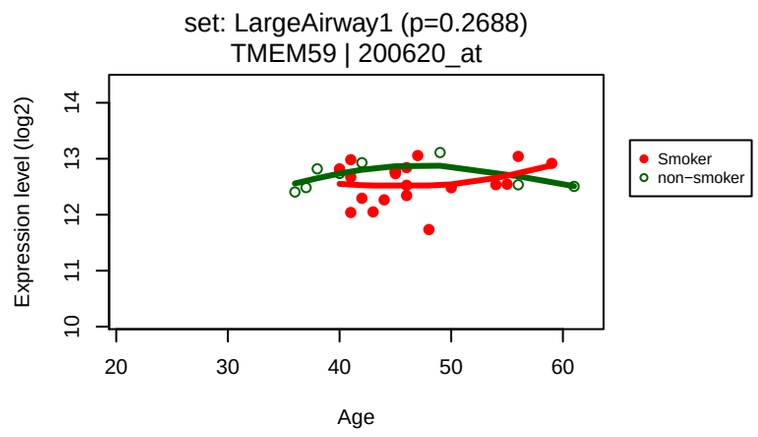

Gene expression of smokers and nonsmokers
gene: TMEM59 | 200620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8905
p=0.9606
p=0.6073
p=0.2688
p=0.0568
p=0.4172
p=0.683
p=0.7564
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

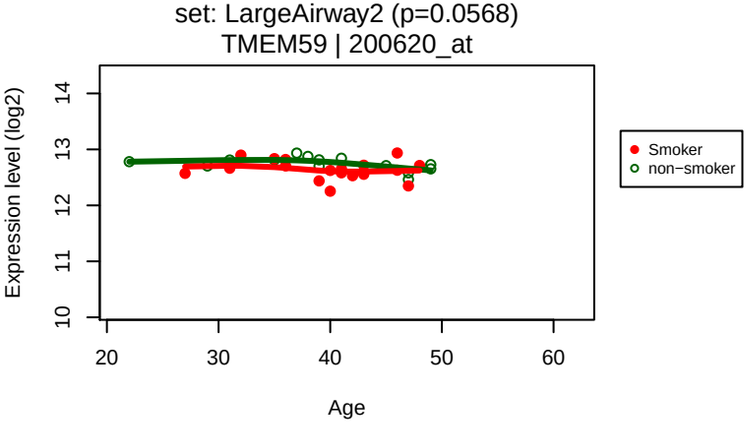

Gene expression ratio (smokers/nonsmokers)
gene: TMEM59 | 200620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8905
p=0.9606
p=0.6073
p=0.2688
p=0.0568
p=0.4172
p=0.683
p=0.7564
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

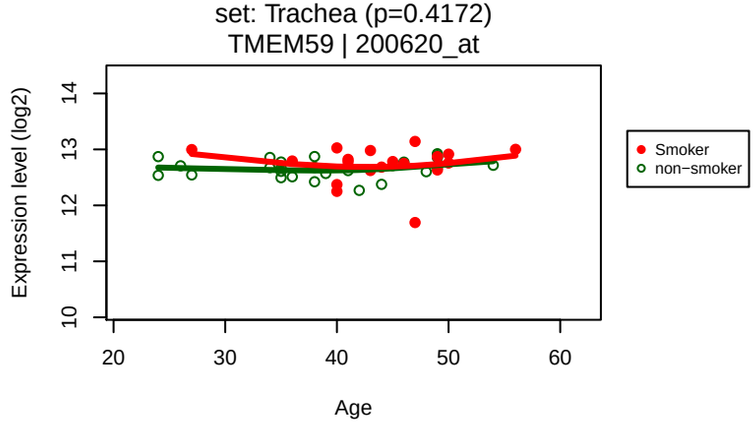

Gene expression ratio in Males
gene: TMEM59 | 200620_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.877
p=0.1449
p=0.3328
p=0.9456
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM59 | 200620_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6641
p=0.2818
p=0.9622
p=0.6749
SmallAirway2
LargeAirway2
Trachea
Macrophage2

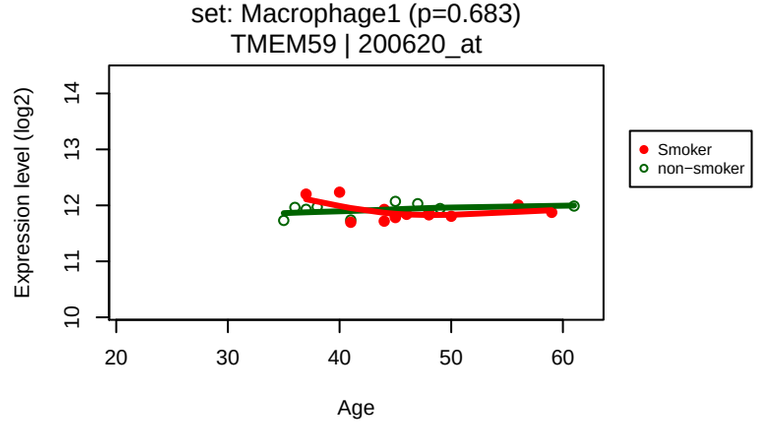

set: SmallAirway1 (p=0.8905)
TMEM59 | 200620_at
Expression level (log2)
Age
Smoker
non−smoker

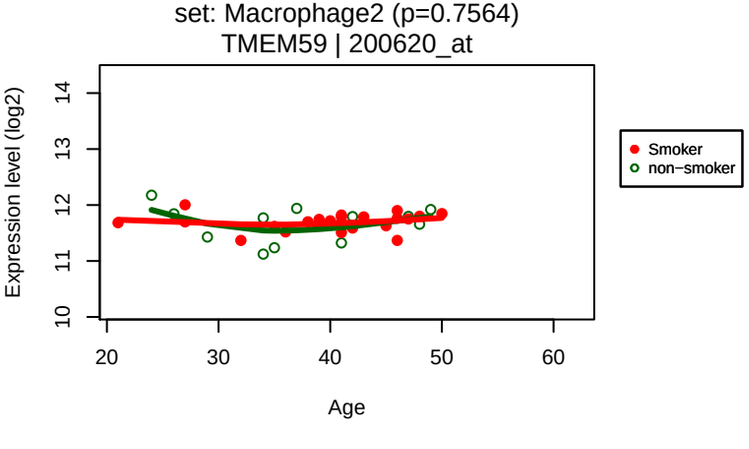

set: SmallAirway2 (p=0.9606)
TMEM59 | 200620_at
Expression level (log2)
Age
Smoker
non−smoker

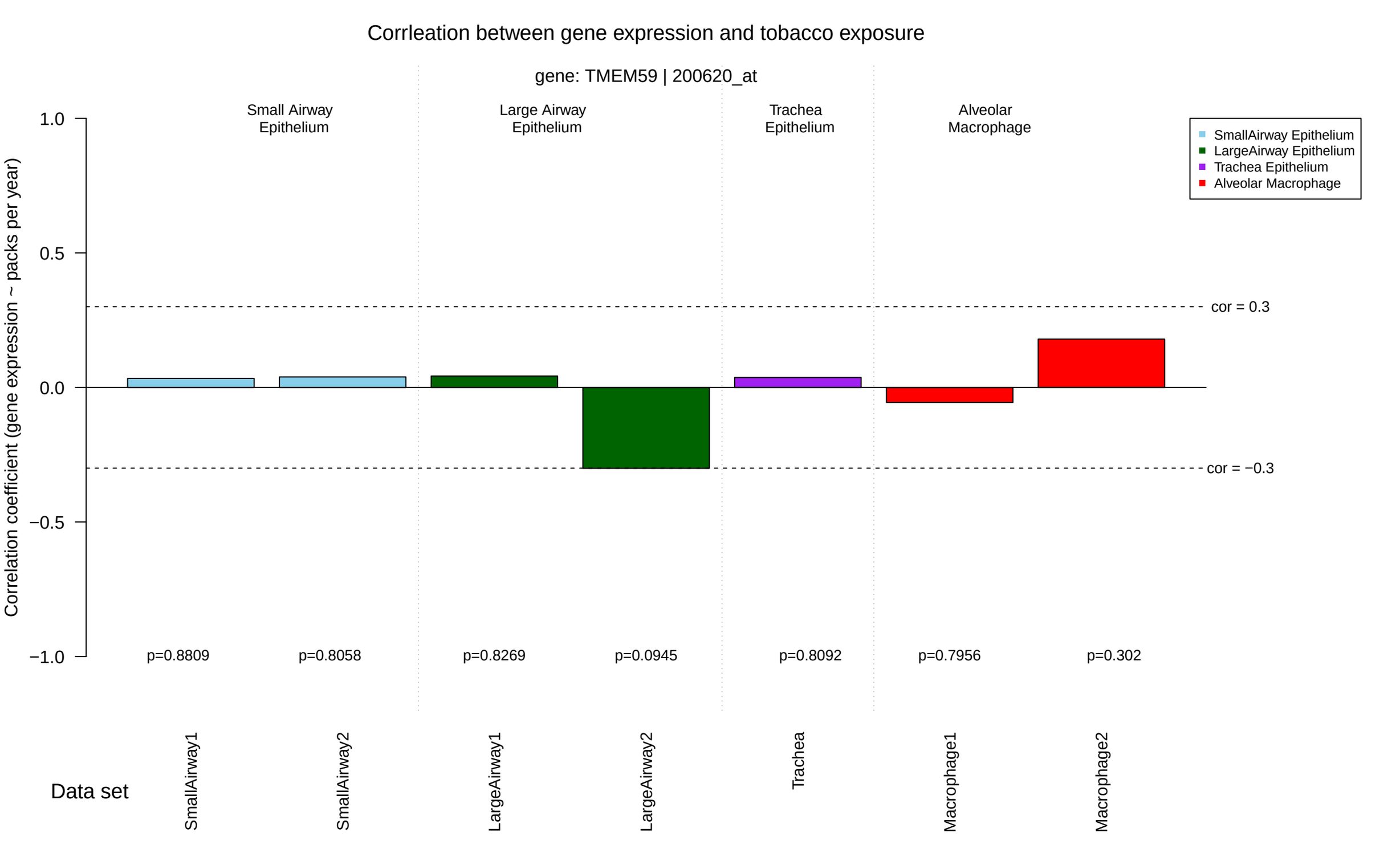

set: LargeAirway1 (p=0.2688)
TMEM59 | 200620_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0568)
TMEM59 | 200620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4172)
TMEM59 | 200620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.683)
TMEM59 | 200620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7564)
TMEM59 | 200620_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM59 | 200620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8809
p=0.8058
p=0.8269
p=0.0945
p=0.8092
p=0.7956
p=0.302
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2